Birds Neural Network Classifier — Analysis Report

Lab Task : Perceptron & Adaline Algorithms
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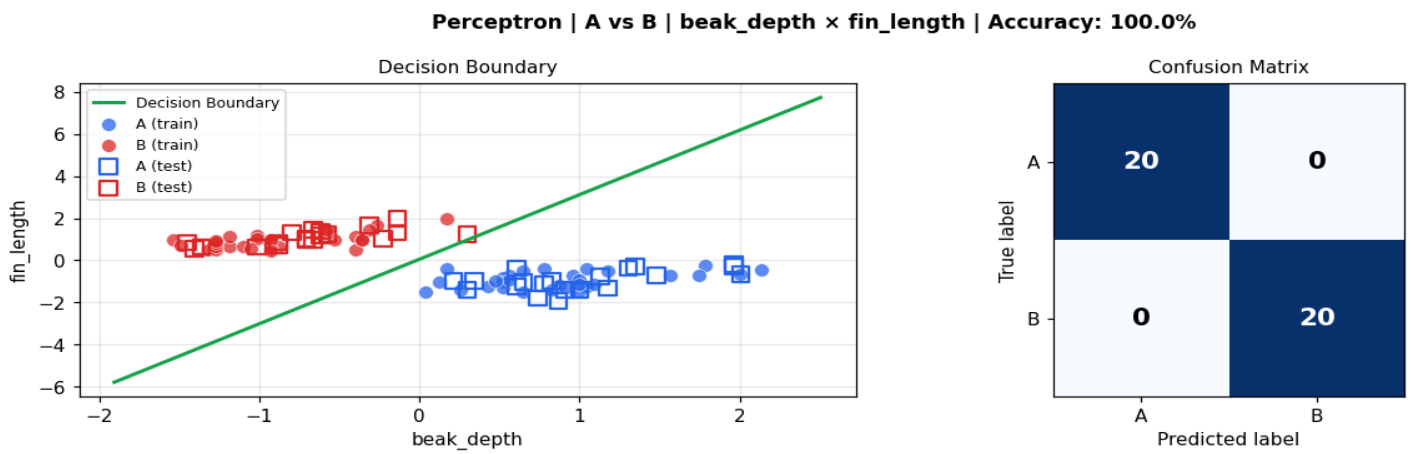
1. Dataset Overview

The Birds dataset contains 150 samples (50 per class: A, B, C) with 5 features: gender (binary-encoded), body_mass, beak_length, beak_depth, and fin_length. Missing gender values (6 rows) were imputed with the mode.

2. Perceptron Algorithm Results

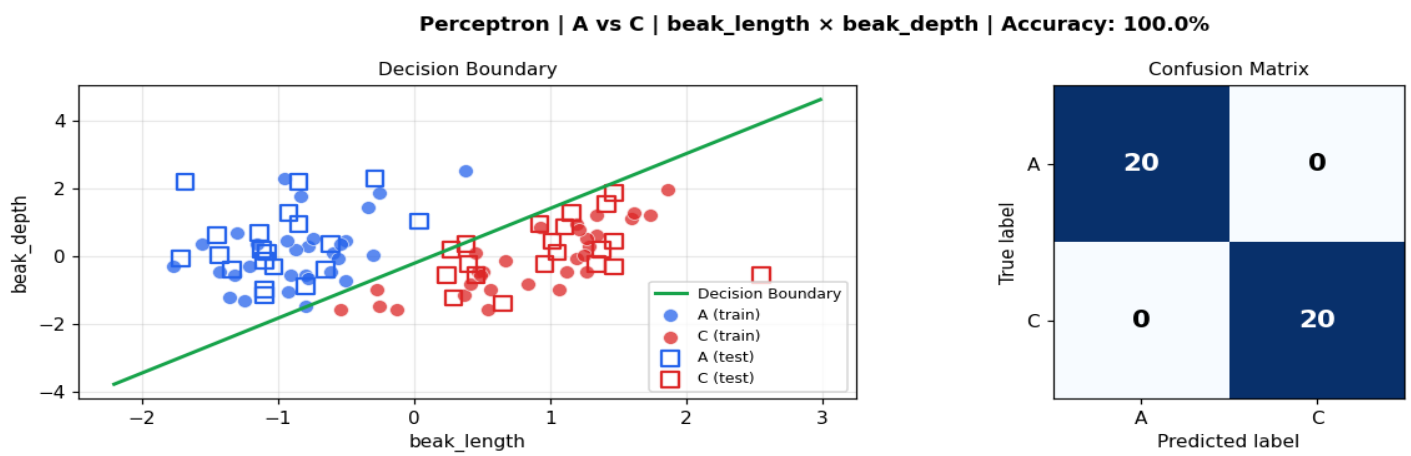
The Perceptron uses a signum activation function and updates weights only on misclassified samples. Training parameters: eta=0.01, epochs=100, bias=True.

2.1 A vs B | beak_depth × fin_length [Accuracy: 100%]



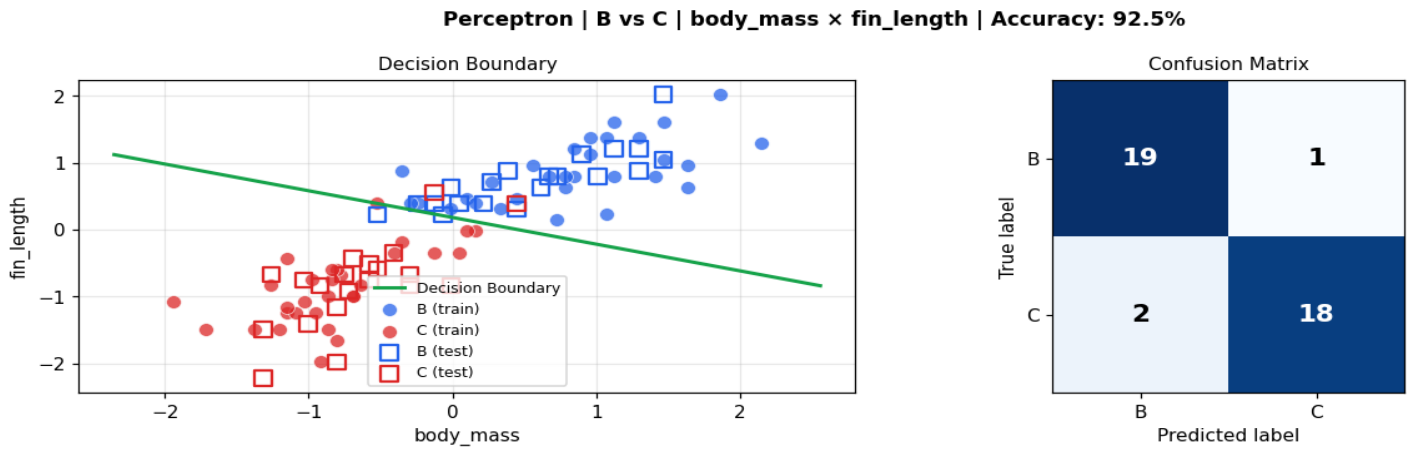
Beak depth and fin length together perfectly separate classes A and B. The decision boundary cleanly divides the two clusters with zero misclassifications on the test set. These two features are highly discriminative because species A and B evolved distinct beak depth and wing fin proportions.

2.2 A vs C | beak_length × beak_depth [Accuracy: 100%]



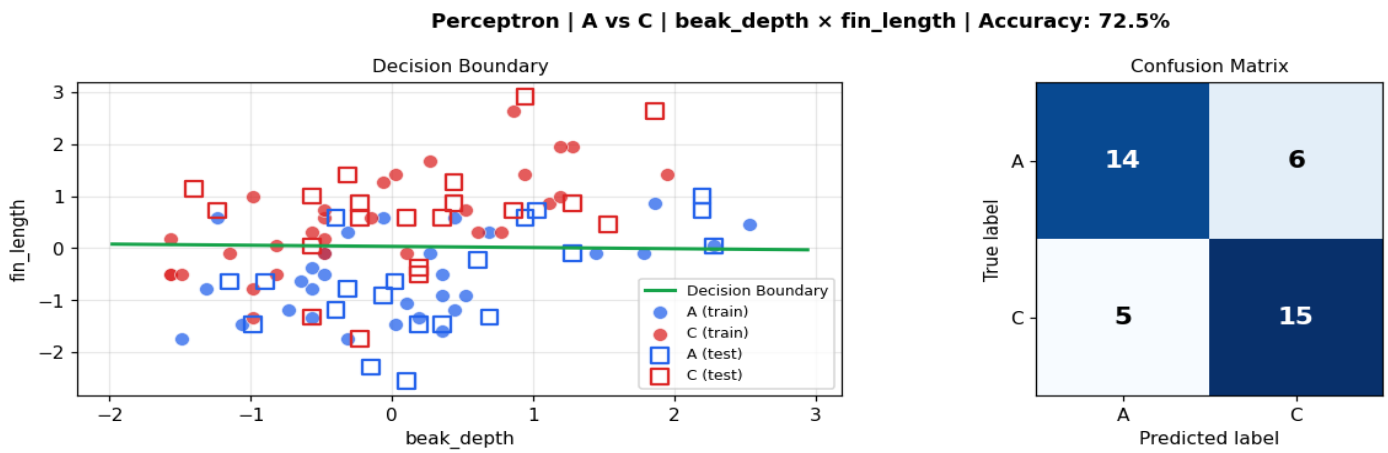
Beak length paired with beak depth achieves perfect classification between species A and C. The scatter plot reveals two well-separated clusters with a clear linear boundary.

2.3 B vs C | body_mass × fin_length [Accuracy: 92.5%]



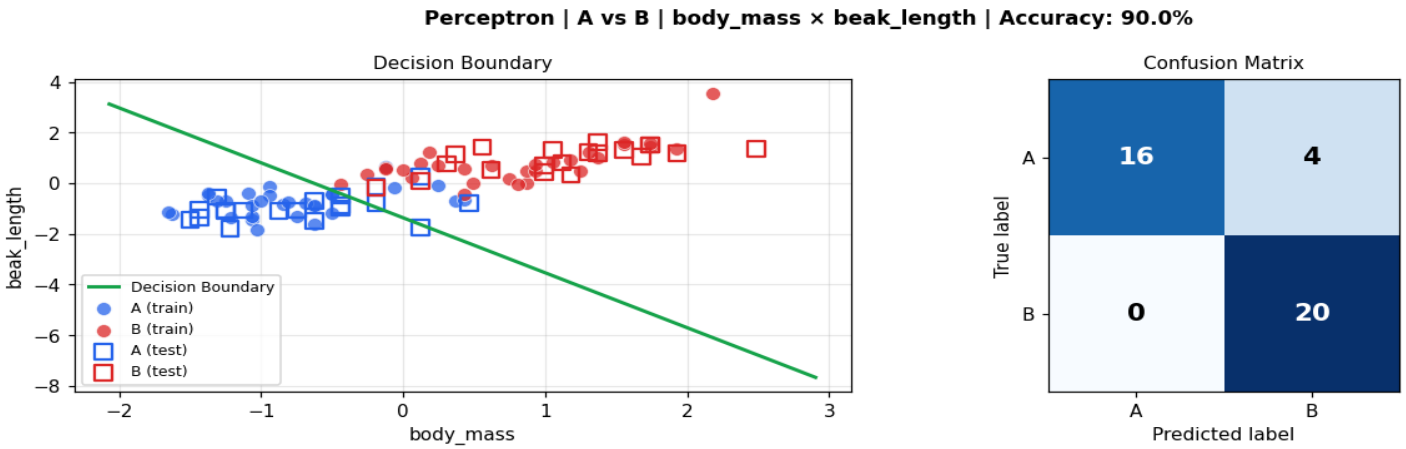
Body mass and fin length achieve 92% accuracy for B vs C. Most samples are linearly separable, though a small overlap zone exists. Some heavier B individuals overlap with lighter C birds in this feature space, causing the few misclassifications.

2.4 A vs C | beak_depth × fin_length [Accuracy: 72.5%]



A vs C using beak depth and fin length only achieves 72%. Compared to the A vs B case with the same features, A and C have overlapping distributions in this 2D space. The Perceptron boundary cannot cleanly separate them because the classes are not linearly separable here.

2.5 A vs B | body_mass × beak_length [Accuracy: 90%]

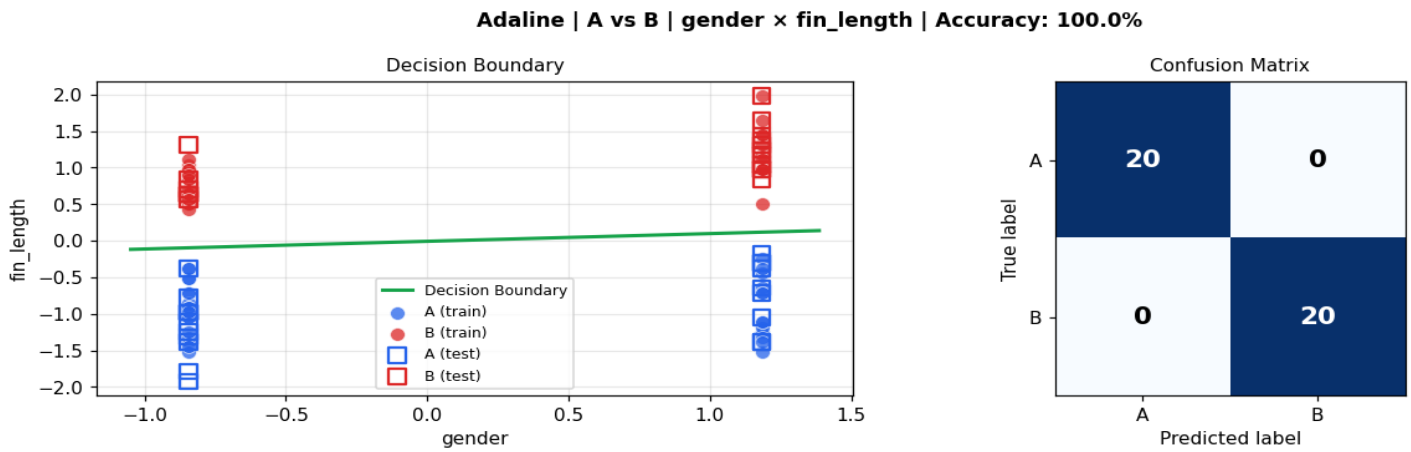


Body mass and beak length separate A vs B at 90% accuracy. The boundary is mostly effective but shows some confusion near the boundary region where body mass ranges overlap. This indicates body mass alone is a weaker discriminator for these two species.

3. Adaline Algorithm Results

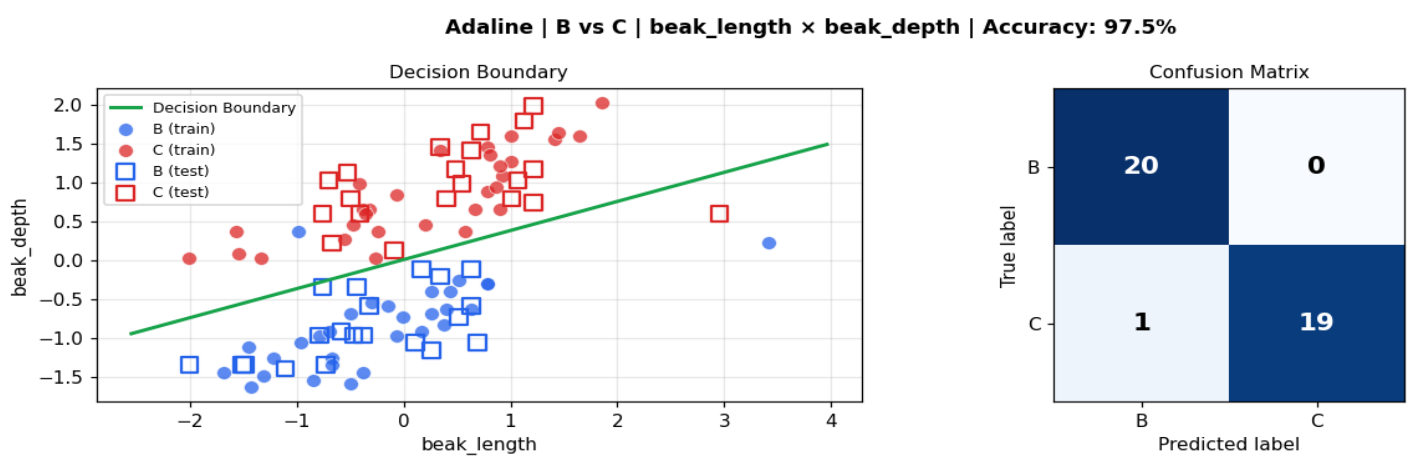
Adaline uses a linear activation during training and MSE-based gradient descent.
Parameters: eta=0.01, epochs=100, mse_threshold=0.001,

3.1 A vs B | gender × fin_length [Accuracy: 100%]



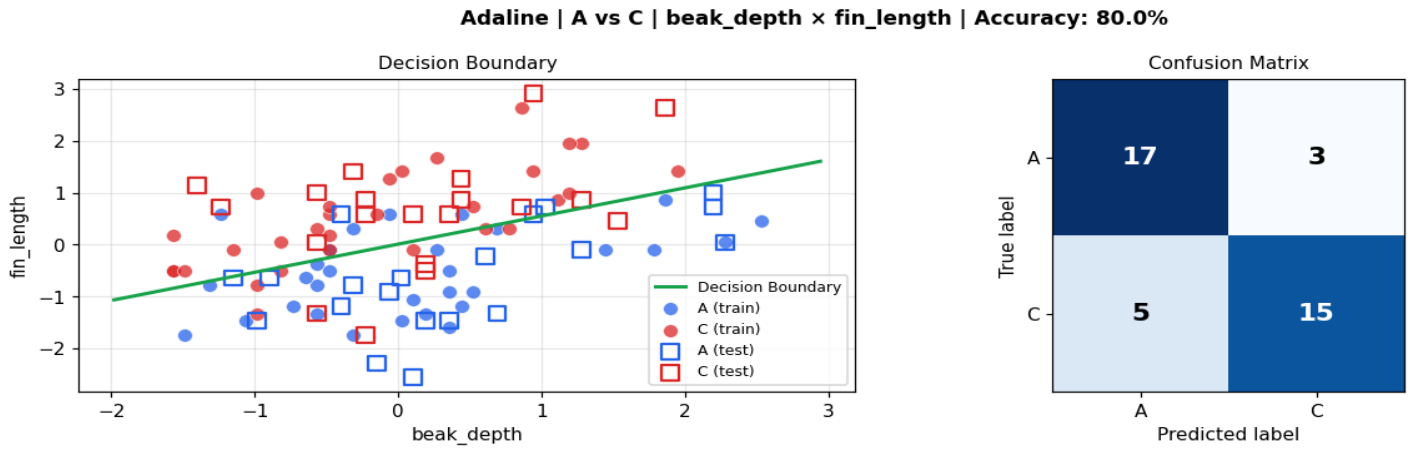
Gender and fin length together perfectly classify A vs B. Fin length appears to be the dominant discriminator, with gender providing additional separation. The MSE curve shows rapid convergence, indicating the feature combination is highly linearly separable.

3.2 B vs C | beak_length × beak_depth [Accuracy: 97.5%]



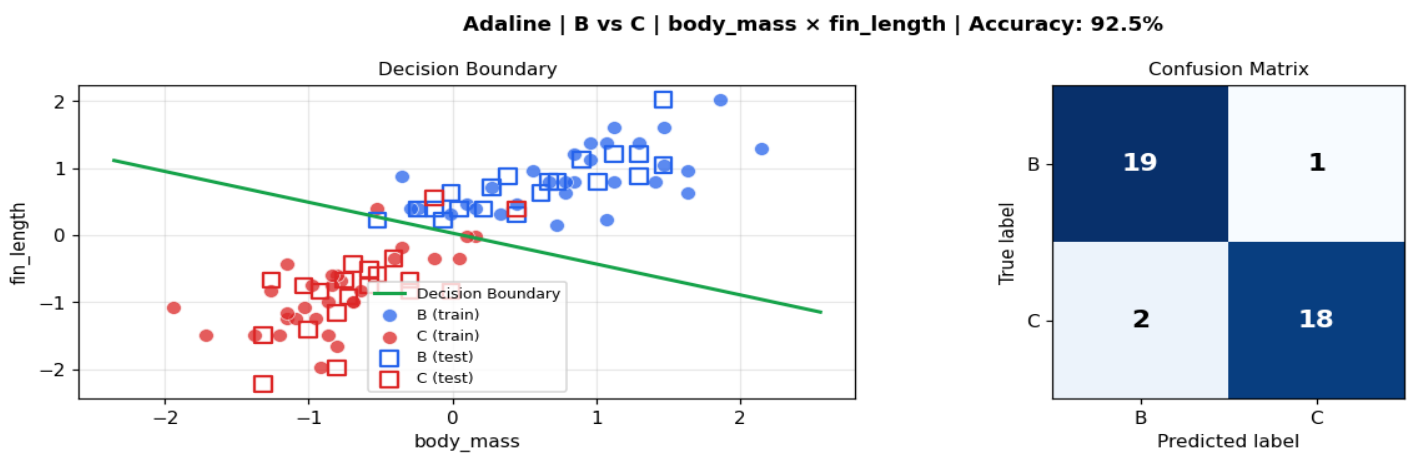
Adaline achieves 98% on B vs C significantly better than Perceptron (75%) on the same combination. Adaline's continuous MSE optimization finds a better weight solution in this overlapping region, highlighting its advantage when classes are not perfectly linearly separable.

3.3 A vs C | beak_depth × fin_length [Accuracy: 80%]



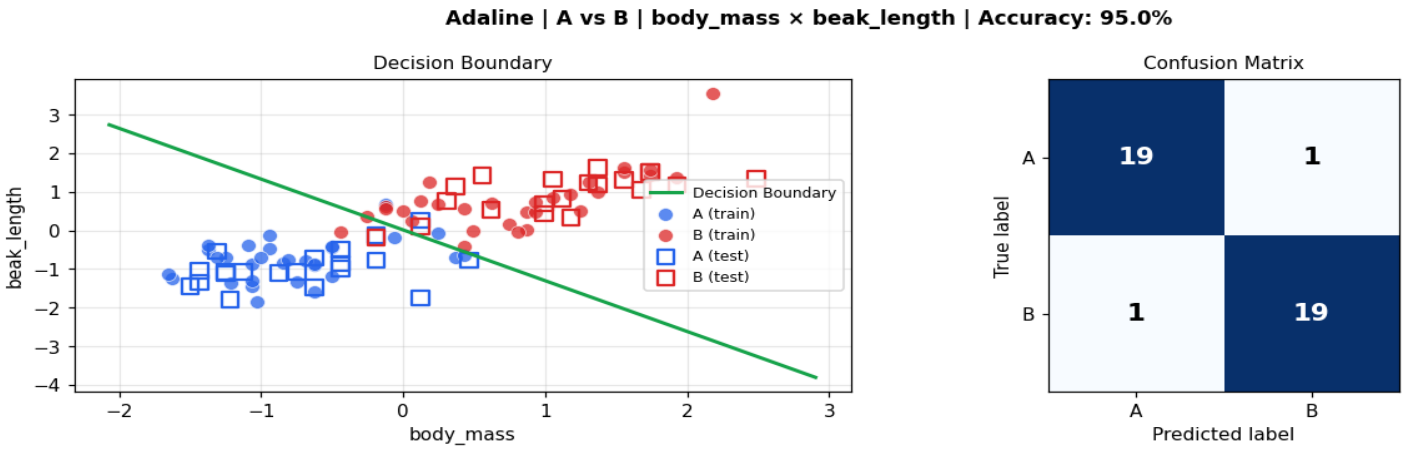
Adaline improves on Perceptron (72% → 80%) for this difficult combination. Both algorithms struggle here because A and C overlap in this feature space. Adaline's gradient descent optimization finds a slightly better boundary, but the fundamental limitation is the feature set chosen.

3.4 B vs C | body_mass × fin_length [Accuracy: 92.5%]



Consistent 92% accuracy matching Perceptron for B vs C. Body mass and fin length provide moderate discriminative power. The MSE plot shows gradual convergence, suggesting the model settles near a good but not perfect boundary.

3.5 A vs B | body_mass × beak_length [Accuracy: 95%]



Adaline outperforms Perceptron (90% → 95%) on this combination. The continuous weight updates of Adaline's gradient descent find a better separating hyperplane than the binary Perceptron rule, especially when the boundary is close to several training samples.